



Manufacturer of Laboratory Instruments and Laboratory Glassware



Materials Enabling **Scientific Discovery**

PRODUCT CATALOGUE

OUR PRODUCT RANGE



GENERAL GLASS APPARATUS



CUSTOMIZED GLASS APPARATUS



PHOTO REACTOR / BIO REACTOR



SOLVENT PURIFICATION SYSTEM



AUTO CLAVE



HIGH-PRESSURE REACTOR



QUALITY
ASSURANCE



CUSTOMIZED
SOLUTIONS



PRECISION
ENGINEERED



DEPENDABLE
PERFORMANCE

Vision and mission

- ❑ To become a leading provider of advanced glass, metal, and ceramic technologies that support scientific research, innovation, and industrial development.
- ❑ To design and manufacture high-quality laboratory equipment and customized scientific systems using advanced glass, metal, and ceramic materials.
- ❑ To continuously improve product quality, safety, and performance through engineering excellence and scientific expertise.

HIGH PRESSURE REACTOR / AUTOCLAVE



RELIABLE



SAFE



EFFICIENT



PRECISE



GMC Technology

INNOVATION | PRECISION | PERFORMANCE

HIGH-PRESSURE REACTOR

Model: GMC-HPR-SG (Split Glove Box Series)

A Split-Type High-Pressure Reactor in which the PID temperature controller and reactor heating assembly are separate units. The heating assembly can be installed inside a glove box or controlled atmosphere enclosure, while the controller remains safely outside, allowing convenient operation without compromising the inert environment. Ideal for air- and moisture-sensitive reactions.

APPLICATIONS

- Hydrogenation
- Catalytic reactions
- Hydrothermal synthesis
- High-pressure chemical processes



Custom-made model
as required by user.



TECHNICAL FEATURES

PARAMETER	SPECIFICATION
Model No.	HPR-SG-25 / HPR-SG-50 / HPR-SG-100 (25, 50, 100 ml)
Maximum Pressure	Up to 150 bar
Maximum Temperature	Up to 200 °C
Reactor Volume	25 ml to 100 ml (As per requirement)
Material	SS316, PTFE
Heating Control	Digital PID Controller
Power Supply	230V AC
Magnetic Stirring	Yes

HIGH-PRESSURE REACTOR

Model: GMC-HPR-HPR-IM (Integrated Magnetic Stirring Series)

A compact, fully integrated **High-Pressure Reactor** with built-in heating assembly and PID temperature controller. The reactor can be placed directly on a laboratory magnetic stirrer, enabling efficient magnetic stirring of the reaction medium while maintaining precise temperature control. Suitable for general laboratory synthesis, hydrothermal, and solvothermal applications.



Custom-made model
as required by user.



TECHNICAL FEATURES

PARAMETER	SPECIFICATION
Model No.	HPR-IM-25 / HPR-IM-50 / HPR-IM-100 (25, 50, 100 ml)
Maximum Pressure	Up to 150 bar
Maximum Temperature	Up to 200 °C
Reactor Volume	25 ml to 100 ml (As per requirement)
Material	SS316, PTFE
Heating Control	Digital PID Controller
Power Supply	230V AC
Magnetic Stirring	Yes

HIGH-PRESSURE REACTOR

Model: HPR-IV (Integrated Vertical Series)

A vertical, fully integrated **High-Pressure Reactor** with built-in heating assembly and PID temperature controller. Its space-saving vertical design is ideal where magnetic stirring is not required. Designed for static high-pressure reactions, crystallization studies, hydrothermal synthesis, and material development with accurate temperature control.



Custom-made model
as required by user.



TECHNICAL FEATURES

PARAMETER	SPECIFICATION
Model No.	HPR-IV-25 / HPR-IV-50 / HPR-IV-100 (25, 50, 100 ml)
Maximum Pressure	Up to 150 bar
Maximum Temperature	Up to 200 °C
Reactor Volume	25 ml to 100 ml (As per requirement)
Material	SS316, PTFE
Heating Control	Digital PID Controller
Power Supply	230V AC
Magnetic Stirring	No

HIGH-PRESSURE AUTOCLAVE

(Model No: GMC-HPA-101)

High-pressure autoclave is a robust reaction vessel designed for conducting pressure-assisted chemical processes. The assembly comprises a stainless-steel reactor body, a pressure gauge, a pressure control valve, and high-pressure fittings to ensure accurate pressure monitoring and secure operation. These reactors enable safe and efficient execution of complex chemical processes such as hydrogenation, oxidation, polymerization, nitration, and catalytic reactions.



Custom-made model
as required by user.



TECHNICAL FEATURES

PARAMETER	SPECIFICATION
Volume	50–100 ml (As per requirement)
Maximum Pressure	Up to 140 bar
Maximum Temperature	Up to 100 °C
Material	SS316

HYDROTHERMAL AUTOCLAVE

(Model No: GMC-HPA-102)



Hydrothermal Autoclave Reactor provides a reliable solution for hydrothermal and solvothermal synthesis under elevated temperature and pressure conditions. The reactor consists of SS outer body with a PTFE-lined reaction chamber, providing excellent corrosion resistance and leak-proof performance. It is available in both Nut & Bolt and Threaded sealing designs with a range of standard reactor volumes to meet diverse laboratory and research requirements. The reactor is ideal for crystal growth, nanomaterial synthesis, catalyst preparation, material digestion, and other high-pressure chemical processes.



Available Standard Specifications are 10ml, 25ml, 50ml, 100ml and customized according to user needs.

TECHNICAL FEATURES

PARAMETER	SPECIFICATION
Maximum Temperature	Up to 200 °C
Maximum working pressure	Up to 100 bar
Material	SS316, PTFE
Sealing Design	Nut-Bolt type

STIRRED-TANK BIOREACTOR

(Model No: GMC-BR-201)

Stirred-Tank Bioreactor is a fermentation system under precisely controlled operating conditions. The reactor is equipped with a top-mounted motor-driven impeller to provide efficient mixing and uniform distribution. Fabricated from high-quality material, the bioreactor incorporates multiple process ports, sampling and addition vessels, and drain valves for convenient operation. It is suitable for microbial fermentation, enzyme production, biotechnology research, process optimization, and pilot-scale bioprocess development.



Custom-made model
as required by user.



TECHNICAL FEATURES

PARAMETER	SPECIFICATION
Volume	As per requirement
Stir Speed	50–100 rpm
Stirring Mechanism	Mechanical Top-Drive Motor
Body and Lid Material	SS316
Weight	40 kg

PHOTOCHEMICAL PRODUCTS



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TRIPLE-JACKETED XENON REACTOR

Model: GMC-PCRX-3L01

Triple-Jacketed Xenon Reactor is suitable for photochemical and photocatalytic reactions under controlled Xenon light irradiation. The assembly comprises an inner quartz immersion well for the Xenon lamp, an intermediate cooling jacket for efficient coolant circulation, and an outer Borosilicate Glass 3.3 reaction vessel for the reaction medium.

- Compatible with UV, visible, and xenon light sources

Available in various capacities with custom configurations



Custom-made model
as required by user.



TECHNICAL FEATURES

PARAMETER	SPECIFICATION
Material Vessel/Lid	Borosilicate Glass 3.3/PTFE
Reactor Bottom	Flat suitable for stirring
Reactor Design	Threaded Outer Borosilicate reactor with Teflon
Temperature	Up to 800 °C

4-LAYER JACKETED UV PHOTOCHEMICAL REACTOR

Model: GMC-PCR-4L02

4-Layer UV Lamp Reactor is designed for laboratory-scale photochemical reactions requiring uniform UV irradiation and efficient temperature control. The reactor consists of a central quartz immersion well for UV lamp, surrounded by concentric cooling and reaction chambers to maximize light utilization while maintaining stable operating conditions. The upper assembly is fitted with a PTFE cover, providing good leak-proof sealing, and secure mounting of the immersion well and process connections. A **gas sparging tube** located at the bottom of the reactor allows the introduction of **nitrogen or other inert gases** for oxygen-free reactions, and continuous purging.

- Compatible with UV, visible, and xenon light sources



Custom-made model
as required by user.



TECHNICAL FEATURES

PARAMETER	SPECIFICATION
Capacity	3L
Reaction Vessel	20 L Borosilicate Glass 3.3 with multiport lid
Material Vessel/Lid	Borosilicate Glass 3.3/PTFE
Operating Temperature	-20°C
Immersion Well	Double jacketed fused silica (Quartz Glass)
Pressure	1 bar
Cooling Jacket	Independent Jacket



4-LAYER JACKETED UV PHOTOCHEMICAL REACTOR

Model: GMC-PCR-20L03

Photocell Reactor consists of an outer Borosilicate Glass 3.3 reactor vessel and an inner double-jacketed quartz immersion well. The quartz immersion well houses UV/LED lamps while maintaining complete isolation from the reaction medium. Independent coolant circulation through the immersion well ensures efficient lamp cooling and maximizes lamp life. The upper assembly is fitted with a **PTFE top cover**, providing leak-proof sealing, and secure mounting of the immersion well and process connections. The reactor incorporates a **bottom gas sparger** for nitrogen or other inert gas circulation. The complete assembly is mounted on a **heavy-duty stainless steel support framework**, ensuring excellent mechanical stability, ease of installation, and convenient operation for pilot-scale and industrial applications.

- Compatible with UV, visible, and xenon light sources



Custom-made model
as required by user.



TECHNICAL FEATURES

PARAMETER	SPECIFICATION
Reaction Vessel	Borosilicate Glass 3.3 Jacketed Vessel
Immersion Well	Jacketed Quartz
Cooling Jacket	Independent Jacket
Support Structure	Structural Steel
Operating Temperature	-23°C (chilled brine)
Operating Pressure	1 bar

5-LAYER JACKETED PHOTOCHEMICAL REACTOR

Model: GMC-PCR-5L04

5-Layer Photochemical Reactor features a central quartz immersion well for housing the UV lamp, surrounded by multiple concentric cooling jackets that ensure effective heat dissipation and uniform irradiation of the reaction medium. Separate inlet and outlet connections provide efficient cooling water circulation, while an additional port allows **nitrogen or other inert gas** purging for oxygen-free reaction conditions.

- Compatible with UV, visible, and xenon light sources

Available in various capacities with custom configurations



Custom-made model
as required by user.



TECHNICAL FEATURES

PARAMETER	SPECIFICATION
Reaction Vessel	Borosilicate Glass 3.3
Immersion well	Double Jacketed Quartz
Operating Temperature	-20 °C
Capacity	1.5L, 3L, 20L
Port	As per requirement



SPECIAL INSTRUMENTAL WORK



CUSTOM SOLUTIONS

Designed and developed to meet specific research needs.



EXPERT TEAM

Skilled professionals delivering precision and reliability.

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SIEVERT APPARATUS

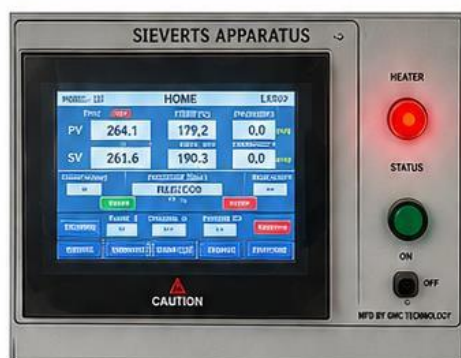
Model: GMC-SA-301

SIEVERT APPARATUS is a robust and precision-engineered manifold system designed for the safe, reliable, and continuous distribution of hydrogen gas.

The distribution panel is equipped with precision pressure gauges, isolation valves, and multiple outlet connections for accurate pressure monitoring and controlled gas delivery to one or more reactors simultaneously.

The complete system is supplied with an integrated digital control panel for precise temperature regulation, and all necessary accessories for seamless operation.

- Custom-made model as required by user



**COMPLETE ASSEMBLY
WITH DISTRIBUTION
SYSTEM AND
CONTROL DISPLAY
PANEL**



TECHNICAL FEATURES

PARAMETER	SPECIFICATION
Material	SS 316
Inlet Connection	As per requirement
Outlet Connection	As per the requirement
Pressure	50 bar
Furnace Temperature Maximum	1050



POLARIMETER TUBE

Model: GMC-PT-401

Stainless Steel (SS) Polarimeter Tube is suitable for precise measurement of optical activity in liquids and solutions. Manufactured with superior-grade stainless steel and high-quality optical components, it provides excellent strength, chemical resistance, and dependable analytical accuracy for laboratory use.

Advantages of SS Body + Quartz Windows

- Higher mechanical strength
- Better pressure resistance
- Better chemical resistance for some solvents
- Safer for industrial applications
- Suitable for vacuum or pressurized operation
- Better durability than glass tubes



TECHNICAL FEATURES

PARAMETER	SPECIFICATION
Material	SS 316
Length/internal Diameter	100 mm/ID-10 mm
End cap	Quartz
Seal	Silicon



HOT AIR OVEN

Model: GMC-HAO-501

Hot Air Oven provides reliable dry-heat sterilization and drying for laboratory glassware, metal instruments, and other heat-resistant materials. It ensures uniform temperature distribution throughout the chamber, enabling effective sterilization under controlled heating conditions. The oven is suitable for sterilizing pipettes, bottles, Petri dishes, test tubes, and laboratory accessories.

Advantages of SS Body + Quartz Windows

- Higher mechanical strength
- Better pressure resistance
- Better chemical resistance for some solvents
- Safer for industrial applications
- Suitable for vacuum or pressurized operation
- Better durability than glass tubes



TECHNICAL FEATURES

PARAMETER	SPECIFICATION
Material	SS316
Chamber Size	one
Voltage	230 Volts
Maximum Temperature	200 °C
Temperature Controller	PID Controller
Type	Hot Air

LONG TUNNEL FURNACE

(long tunnel 1.5 Metre)
(Model No: GMC-LTF-601)

Long Tunnel Furnace provides a controlled heating environment for continuous thermal processing of materials at temperatures up to 800 °C. The furnace features a robust steel outer body and a refractory-lined heating chamber insulated with high-grade ceramic fiber to ensure excellent thermal efficiency and uniform temperature distribution. Straight Kanthal heating elements provide reliable and consistent heating throughout the chamber, while the digital PID temperature controller enables precise temperature regulation.

- The long tunnel furnace has been designed to simulate an industrial-scale continuous heating protocol while reducing energy losses.
- *Available in various capacities with custom configurations*

Technical features

Parameter	Specification
Operating Temperature	800°C
Maximum Temperature	1050°C
Heater	Kenthol
Material and Insulation	SS316 and Ceramic wool





CUSTOMIZED-GLASSWARE WORK

Precision | Quality | Custom Solutions



CUSTOM SOLUTIONS

Tailored glassware as per your specific requirements.



PREMIUM QUALITY

Made from high borosilicate glass for excellent chemical and thermal resistance.



PRECISION CRAFTED

Engineered for accuracy, consistency, and long-lasting performance.



WIDE APPLICATION

Ideal for research, industrial, and laboratory applications.

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DEO Assembly (Model No: GMC-DEO-701)



The advanced DEO System is an industrial processing system for efficient deodorization, purification, and treatment of industrial oils. The system effectively removes odor, moisture, volatile impurities, and dissolved gases to improve oil quality and performance. Suitable for edible oils, transformer oils, lubricants, and specialty oil applications.

- **Applications:** Edible oil processing, transformer oil treatment, lubricating oil purification, refinery and chemical process industries.
- **Optional Features:** Vacuum system, condenser unit, PLC automation, skid-mounted design.

Technical features

Parameter	Specification
Material	Borosilicate glass 3.3
Joints	As per requirement
Size/Dimension	As per requirement

SPECIAL GLASSWARE



4-NECK CYLINDRICAL FLASK WITH CONDENSERS

(Product code: GMC-GG-4NCF-101)

Reaction Flask Assembly comprises a multi-neck borosilicate glass reaction flask with interchangeable water-cooled condensers for laboratory-scale synthesis, reflux, and distillation applications. The modular configuration accommodates a variety of experimental requirements.

4-NECK CYLINDRICAL FLASK WITHOUT CONDENSERS

(Product code: GMC-4NCF-02)

- Available in various capacities with custom configurations



SOLVENT STILL HEAD

(Product code: GMC-GG-SSH-102)

Solvent Distillation Head is a laboratory glass assembly for the safe distillation, collection, and controlled transfer of solvents. Fabricated from **Borosilicate Glass 3.3**, the assembly incorporates standard ground-glass joints and PTFE stopcocks for controlled solvent dispensing and efficient operation.

- Available in various capacities with custom configurations



SPECIAL GLASSWARE-REFLEX DIVIDER

Product code: GMC-SG-RD-702



The **Glass Reflex Divider** is manufactured from Borosilicate Glass 3.3 for accurate control of condensate distribution during laboratory reflux and fractional distillation processes. It offers excellent thermal stability, high optical clarity for process observation, and outstanding chemical resistance against most laboratory reagents. The smooth, polished glass surface ensures easy cleaning and reliable operation, while the divider can be supplied in flat or custom configurations to suit specific process requirements.

Technical Features

Parameter	Specification
Material	Borosilicate glass 3.3
Joints	As per requirement
Valve	PTFE, lateral, adjustable
Size/Dimension	As per requirement
Operating Temperature	Up to 250–300°C



DISTILLATION HEAD WITH REFLUX DEWAR CONDENSER



DISTILLATION HEAD WITH REFLUX

(Product code: GMC-SG-DH-703)

A compact **Glass Distillation Head** incorporating an integral Vigreux vapor path and reflux condenser for efficient fractionation and controlled reflux during distillation. The assembly includes dedicated coolant inlet/outlet connections, a for process monitoring. The lower outlet provides controlled collection and distribution of condensate, making it suitable for laboratory-scale distillation and solvent purification setups.



DEWAR CONDENSER

(Product code: GMC-SG-DW-704)

Dewar Condenser feature a wide top opening for convenient charging of cooling media such as dry ice, liquid nitrogen, or dry ice–acetone mixtures. Typically manufactured from borosilicate glass, they provide good thermal shock resistance and excellent chemical resistance, making them suitable for low-temperature condensation applications.



REFLEX DIVIDER

(Product code: GMC-SG-RD-705)



Reflex Divider is suitable for regulating the distribution of condensed liquid during reflux and distillation operations. Its modular design allows seamless integration with laboratory glass assemblies for applications requiring adjustable reflux conditions. The reflux divider is available in different joint sizes and can be customized to meet specific process requirements.

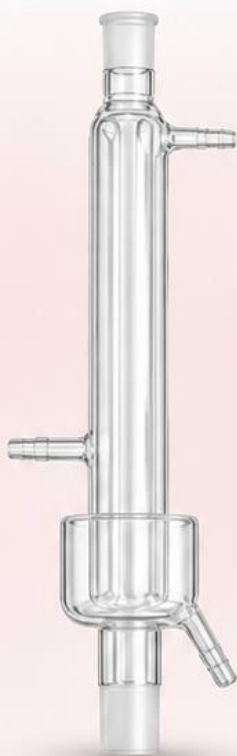
SPECIFICATION

- **Top angled female ground-glass joint-** for connecting an addition funnel, condenser, thermometer adapter, or gas inlet.
- **Horizontal main body with male and female ground-glass joints-**connects between the reaction vessel and the upper glassware.
- **Central pressure-equalizing chamber-**helps maintain pressure equilibrium during reagent addition.
- **Bottom male ground-glass joint-**fits into the reactor or reaction kettle cover.

TECHNICAL FEATURES

Parameter	Specification
Material	Borosilicate glass 3.3/PTFE stopcock
Joints	As per requirement
Capacity	50 ml, 100ml
Size/Dimension	As per requirement

DOUBLE-JACKETED CONDENSER



DOUBLE-JACKETED CUP CONDENSER

(Product code: GMC-SG-JC-706)

Cup Condenser comprises a Borosilicate Glass 3.3 condenser with an integrated collection cup for reflux reaction. The collection cup prevents condensation or coolant leakage from reaching the heating bath or surrounding work area, while the condenser ensures efficient heat transfer with chilled water or low-temperature circulating fluids.



DOUBLE-JACKETED CONDENSER WITHOUT HOLD-UP

(Product code: GMC-JC-707)

Condenser Without Hold-Up is specially designed to eliminate liquid hold up during reflux and distillation processes. Its continuous inner tube extends directly to the outlet joint, minimizing dead volume and ensuring complete condensate drainage. This design reduces product loss, prevents residual solvent or reaction mixture from remaining inside the condenser, and facilitates easier cleaning between experiments.

REACTION KETTLE ASSEMBLY

(Product code: GMC-RKA-801)



Reaction Kettle Assemblies are modular laboratory systems for chemical synthesis, photochemical reactions, catalysis, and process development. The modular construction comprises interchangeable kettle bodies and multi-neck kettle covers, allowing flexible installation of condensers, mechanical stirrers, thermometers, gas inlet/outlet adapters, and addition funnels. The assemblies are secured using **stainless steel quick-release clamps** and **PTFE sealing gaskets** to ensure a leak-tight seal during operation. The reaction kettles are available in standard and bottom-discharge configurations to facilitate efficient charging, reaction, and product recovery.

SPECIFICATIONS

Parameter	Specification
Material	Borosilicate glass 3.3
Locking System	SS clamps/PTFE sealing gaskets
Temperature	Up to 100 °C
Size and Dimension	As per requirement



ALUMINUM AND CERAMIC WOOL-INSULATED VACUUM COLUMN

(Product code: GMC-IVC-901)

Step-1

Knitt mesh (High flux SS) packing
inside column



Step 2

Vacuum insulated 1200 mm column



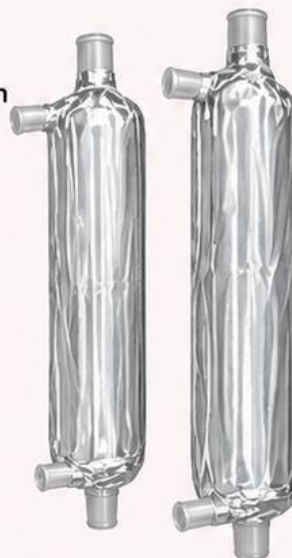
Step 3

Ceramic wool packing



Step 4

Aluminum sheet insulation



The **Customized Vacuum-Insulated Distillation Column** incorporates knitted SS-wire mesh packing to enhance vapor-liquid contact, while the evacuated vacuum jacket provides effective thermal insulation. An additional layer of ceramic wool insulation enclosed within an aluminum protective cover further reduces heat loss and improves temperature stability throughout the distillation process.

- *Available in custom configurations*



SCHLENK LINES

(Product code: GMC-SL-1001)



Schlenk Line Assembly is a precision-engineered system for vacuum and inert gas use to conduct reactions under oxygen- and moisture-free conditions. The multi-port design enables efficient evacuation, inert-gas purging, and parallel operation of multiple glass assemblies, ensuring safe and reproducible handling of sensitive materials.

SPECIFICATIONS

Parameter	Specification
Material	Borosilicate glass 3.3
No. of Ports	3-5
Tube length	As per requirement
Tube connectors	B19/B24



GENERAL GLASSWARE

SCHLENK PRESSURE TUBE

(Product code: GMC-GG-103)



Schlenk Tube is designed for handling air- and moisture-sensitive compounds under vacuum or inert atmosphere. Manufactured from **Borosilicate Glass 3.3**, the assembly incorporates a PTFE high-vacuum valve and a sidearm for gas or vacuum connection.

Size and Dimension: As per requirement

STORAGE BOTTLE WITH GLASS JOINT

(Product code: GMC-GG-104)



20 ml Glass Storage Bottle with Ground Joint is designed for the safe storage and handling of laboratory reagents, reference standards, and moisture-sensitive samples. Manufactured from **Borosilicate Glass 3.3**, it features a precision-ground glass joint for a secure, leak-resistant seal.

Size and Dimension: As per requirement

SPECIFICATIONS

Parameter	Specification
Material	Borosilicate glass 3.3
No. of Ports	Sidearm + Vacuum connection
Tube length	As per requirement
Tube connectors	B19/B24

Parameter	Specification
Material	Borosilicate glass 3.3
Capacity	20 ml
Joint type	Ground joint
Size/Dimension	As per requirement